

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I P. Greethona (192420017) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS, DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms to 40 ms**, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning

Video Conferencing TCP / UDP and Application Layer:

1. Analyze whether the problem is primarily caused by the transport protocol or application layer.

1. The problem is mainly caused by network conditions and transport protocol behavior.

2. The network has 8% packet loss and 180ms jitter, which are very high for real-time communication.

3. TCP retransmits lost packets which can cause delays and head of line blocking.

4. UDP avoids retransmission delays but does not guarantee delivery.

5. Application level buffering can further increase delay if the buffer is too large.

2. Why is this important for real-time multimedia?

1. Video conferencing requires low latency.

2. It requires low jitter for smooth audio and video.

3. Small packet losses are generally preferable to delayed packets.

4. TCP retransmission can cause old packets to arrive late.

5. UDP is more suitable for real-time media because it avoids retransmission delays.

Recommended improvements:-

1. Use UDP-based retransmission real-time communication such as RTP/RTCP
2. Apply adaptive bitrate control.
3. Use adaptive jitter buffering.
4. Apply QoS to prioritize audio/video traffic.
5. Use forward Error Correction (FEC) for packet loss.

2. Global DNS INFRASTRUCTURE

4. Reasons for high DNS resolution time.

1. DNS requests may travel to geographically distant servers
2. A centralized DNS server may become overloaded.
3. Multiple DNS resolution steps increase latency
4. DNS cache misses require additional queries
5. DNS architecture to reduce lookup time.

1. Deploy Anycast DNS servers in multiple geographical regions

2. Use nearby DNS servers in for users in different continents

3. Enable DNS caching at recursive solvers

4. Use DNS pre-fetching for frequently accessed domains

Advantages and limitations

- 1. Lower DNS lookup latency
- 2. Better global performance
- 3. Higher availability
- 4. Improved Accuracy

3. Distributed DNS for cloud services

7. Impact of centralized DNS architecture.

- 1. All DNS queries are handled by one primary server
- 2. Heavy traffic can overload the service.
- 3. DNS response time increases significantly
- 4. The reported 800ms lookup time delays application access

8. Distributed DNS solution

- 1. Deploy DNS servers in multiple geographical regions
- 2. Use Anycast routing to distribute DNS requests
- 3. Deploy multiple authoritative DNS servers.
- 4. Use recursive DNS resolvers close to users

9. TTL, Anycast and DNS caching

- 1. Determines how long DNS records remain cached
- 2. Uses the same IP address for multiple DNS servers
- 3. Stores previously resolved DNS records.

4. High frequency stock Trading and Tcp

10. Three Tcp factors causing latency

1. Flow control.

1. prevents the sender from overwhelming the receiver

2. uses the receiver's advertised window

3. A small window restricts data.

2. Nagle's Algorithm

1. Combines small Tcp packets into larger packets

2. Reduces network overhead.

3. SYN queue limitations

1. Tcp uses a three way handshake to establish connections

2. large numbers of simultaneous connections

11. Effect on high-frequency systems

1. flow control can restrict the rate.

2. Small Tcp windows can increase waiting time

3. Nagle's Algorithm can delay

12. Recommended Tcp configuration changes

1. Disable Nagle's Algorithm using `Tcp_NODELAY` for latency-sensitive messages

2. Use persistent ~~Tcp~~ connections

3. Monitor Tcp retransmissions.